

POLITECNICO
MILANO 1863

Best Bike Paths – RASD Document

RASD document for the
Software Engineering 2 course assignment

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Contents

Contents **i**

1 Introduction **1**
1.1 Purpose . . . 1
1.1.1 Goals . . . 1
1.2 Scope . . . 2
1.2.1 World Phenomena . . . 2
1.2.2 Machine Phenomena . . . 3
1.2.3 Shared Phenomena . . . 3
1.3 Definitions, Acronyms, Abbreviations . . . 4
1.4 Reference Documents . . . 4
1.5 Document Structure . . . 4

2 Overall Description **5**

3 Specific Requirements **7**
3.1 Functional Requirements . . . 7
3.1.1 Requirements . . . 7
3.1.2 Use Cases Diagrams . . . 9
3.1.3 Use Cases . . . 9
3.1.4 Mapping on Goals . . . 28
3.2 Performance Requirements . . . 32
3.3 Design Constraints . . . 34
3.3.1 Standard compliance . . . 34
3.3.2 Hardware limitations . . . 34
3.4 Software System Attributes . . . 35
3.4.1 Reliability . . . 35
3.4.2 Availability . . . 35
3.4.3 Security . . . 35
3.4.4 Maintainability . . . 36

3.4.5 Portability 36

4 Formal Analysis Using Alloy **37**

4.1 Signatures 37

4.1.1 Status 37

4.1.2 Segment 37

4.1.3 Path 38

4.1.4 Statistics 38

4.1.5 Trip 38

4.1.6 RegisteredUser 39

4.2 Facts 39

4.2.1 Segment facts 40

4.2.2 Path facts 40

5 Effort Spent **41**

6 References **43**

List of Figures **45**

List of Tables **47**

1 Introduction

1.1. Purpose

The primary purpose of this project, initiated by a leading Bike Association, is to develop and deploy an advanced support system for the cycling community designed to thrive within a transport landscape dominated by cars.

The core idea is to foster a collaborative digital environment where cyclists can actively record, consult, and publish both their own and other users' riding paths.

This platform will serve as the definitive centralized resource for finding highly recommended routes tailored precisely to the user's specific experience, preferences, and safety considerations.

Ultimately, the system aims to significantly enhance the quality of the cycling experience by providing verified, community-rated routes, thereby connecting riders and making high-quality path discovery universally accessible

1.1.1. Goals

Here are the goals that BBP has to fulfill. BBP has to:

Goal	Description
G1	Recognize suitable bike paths
G2	Allow the user to browse different bike paths
G3	Allow the user to see bike paths statistics
G4	Allow the user to search for paths between two locations
G5	Allow the user to see paths optimality based on other registered users' reviews
G6	Allow user to see weather information when available
G7	Allow users to register to the application using credentials
G8	Allow registered user to review published bike paths
G9	Allow registered user to confirm or reject false positives in track recording
G10	Allow registered user to let the system automatically record suitable bike paths
G11	Allow registered user to manually create suitable bike paths
G12	Allow registered user to publish bike paths
G13	Allow the store of bike paths data for registered users
G14	Allow registered user to add publishable information about bike paths

Table 1.1: List of Goals

1.2. Scope

1.2.1. World Phenomena

WP	Description
WP1	A user physically moves through space using a bicycle
WP2	The physical conditions of a bike path or street change (e.g., presence of potholes, road maintenance status, or obstacles)
WP3	The meteorological conditions in the geographic area vary (e.g., temperature, wind speed)
WP4	A user intends to reach a specific destination starting from a specific origin point

Table 1.2: World Phenomena

1.2.2. Machine Phenomena

MP	Description
MP1	The system guesses that the user is biking by analyzing the detected speed
MP2	The system computes statistics related to a trip (e.g., total distance covered, average speed, performance metrics)
MP3	The system identifies potential anomalies (e.g., potholes) by processing raw data from the device sensors (accelerometer and gyroscope)
MP4	The system computes the score of a path based on its status and its effectiveness in reaching the destination
MP5	The system merges path information incoming from multiple users, evaluating data freshness and the number of confirmations received
MP6	The system stores trip history and path information in the database for future retrieval

Table 1.3: Machine Phenomena

1.2.3. Shared Phenomena

World-Controlled Phenomena

SP_WC	Description
SP_WC1	The user sends a command to start or stop recording a trip
SP_WC2	The mobile device sensors (GPS, accelerometer, gyroscope) transmit position and movement data to the system
SP_WC3	The user manually inserts data regarding a street (e.g., street name, status)
SP_WC4	The user confirms or corrects information automatically detected by the system (e.g., confirming a pothole)
SP_WC5	The user inputs search parameters for a path (origin and destination)
SP_WC6	The external weather service transmits current condition data (e.g., temperature, wind) to the system

Table 1.4: Shared Phenomena – World-Controlled

Machine-Controlled Phenomena

SP_MC	Description
SP_MC1	The system visualizes available bike paths between two points on a map, highlighting those with better scores
SP_MC2	The system displays trip statistics to the user (speed, distance, etc.)
SP_MC3	The system presents a confirmation request to the user regarding automatically detected information

Table 1.5: Shared Phenomena – Machine-Controlled

1.3. Definitions, Acronyms, Abbreviations

- BBP: Bike Buddy Project
- RASD: Requirements Analysis and Specification Document
- WP: World Phenomenon
- MP: Machine Phenomenon
- SP_WC: Shared Phenomenon - World Controlled
- SP_MC: Shared Phenomenon - Machine Controlled

1.4. Reference Documents

- Specification document assignment
- IEEE standard documentation for RASD
- Materials for the Software Engineering 2 course made available on WeBeep

1.5. Document Structure

2 | Overall Description

Chapter 2

3 Specific Requirements

3.1. Functional Requirements

3.1.1. Requirements

ID	Description
R1	The system shall allow new users to register an account
R2	The system shall allow user not registered to access without an account if they don't want to register
R3	The system shall allow registered users to log in to the application
R4	The system shall allow registered users to view and update their personal profile information
R5	The system shall allow users to log out of their current session
R6	The system shall allow registered users to record their cycling trips
R7	The system shall store recorded trips to maintain a history of the user's cycling activities
R8	The system shall compute and display total distance covered for each trip
R9	The system shall compute and display average speed for each trip
R10	The system shall compute and display total time spent speed for each trip
R11	The system shall compute and display elevation gain for each trip
R12	The system shall compute and display calories spent for each trip
R13	The system shall compute and display additional performance metrics for the trips
R14	The system shall retrieve meteorological data (weather conditions, temperature, wind speed) from an external service and enrich the trip data when available
R15	The system shall allow registered users to insert information about bike paths
R16	The system shall allow registered users to make publishable information about bike paths

Continue in the next page...

ID	Description
R17	The system shall allow registered users to add info about the status of a bike path
R18	The system shall allow registered users to add the presence of relevant obstacles on the bike path
R19	The system shall recognise if the users are cycling on an exists bike path
R20	The system shall recognise if the users are cycling on a path where cars are rare and speed limits under 50kmh
R21	The system shall calculate the distance of a path
R22	The system shall allow registered users to manually insert bike path data by specifying the names of the streets included in the path
R23	The system shall allow registered users to manually specify the status of the street
R24	The system shall allow users to manually report the presence of obstacles on the street
R25	The system shall automatically infer that the registered user is cycling based on their speed
R26	The system shall collect GPS data to reconstruct the path followed by the user during the ride
R27	The system shall acquire data from the mobile device's accelerometer and gyroscope to detect basic metrics
R28	The system shall analyze sensor data to identify potential path irregularities
R29	The system shall require the user to confirm or correct the automatically acquired data
R30	The system shall allow registered users to mark their inserted data as "publishable" to make it available to the community
R31	The system shall merge information about the same path from multiple users
R32	The system shall prioritize data based on timeframe of execution and the number of confirmations to determine the current status
R33	The system shall allow any user to search for bike paths by specifying an origin and a destination
R34	The system shall find for its data bike paths where an origin and a destination are specified by any users
R35	The system shall visualize the bike paths connecting the specified origin and destination on a map
R36	The system shall compute a score for each path based on its status and its effectiveness in reaching the destination from the origin

Continue in the next page...

ID	Description
R37	If multiple paths exist, from the user's request the system shall display them sorted by their calculated score

Table 3.1: Requirements List

3.1.2. Use Cases Diagrams

3.1.3. Use Cases

In this section, they are explained and represented the main identified use cases. For each Use Case, the documentation follows a two-step structure:

- 1. Use Case Table: A structured table detailing specific fields
- 2. Diagram: A Sequence Diagram visualizing the steps described in the table

UC1. Sign in to BBP

Actor	abc
Entry conditions	abc
Event Flow	• abc
Exit condition	abc
Exceptions	• abc

Table 3.2: Sign in to BBP

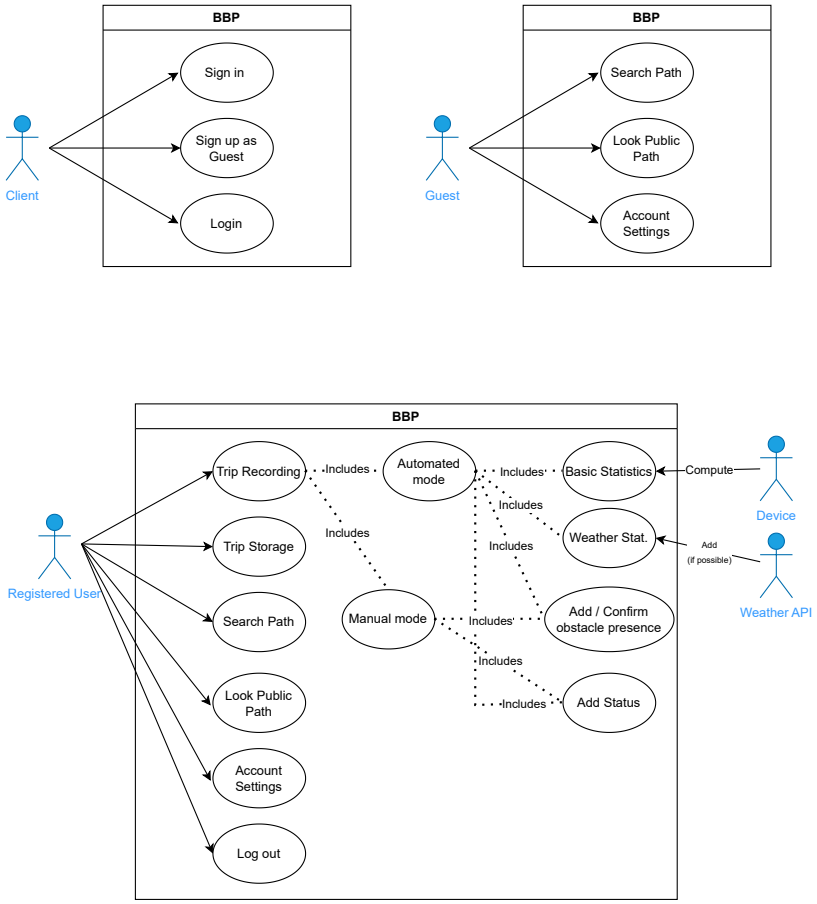


Figure 3.1: Use Cases

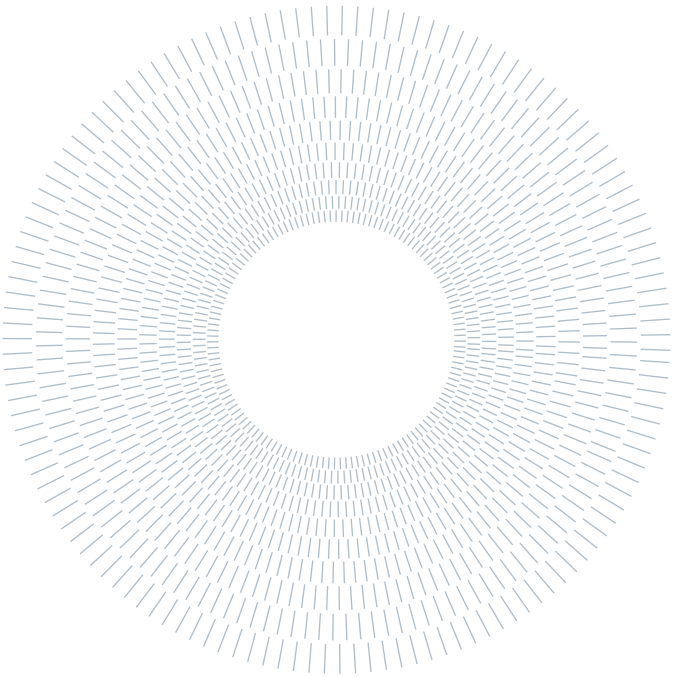


Figure 3.2: Sign in to BBP

UC2. Login to BBP

Actor	abc
Entry conditions	abc
Event Flow	• abc
Exit condition	abc
Exceptions	• abc

Table 3.3: Login to BBP

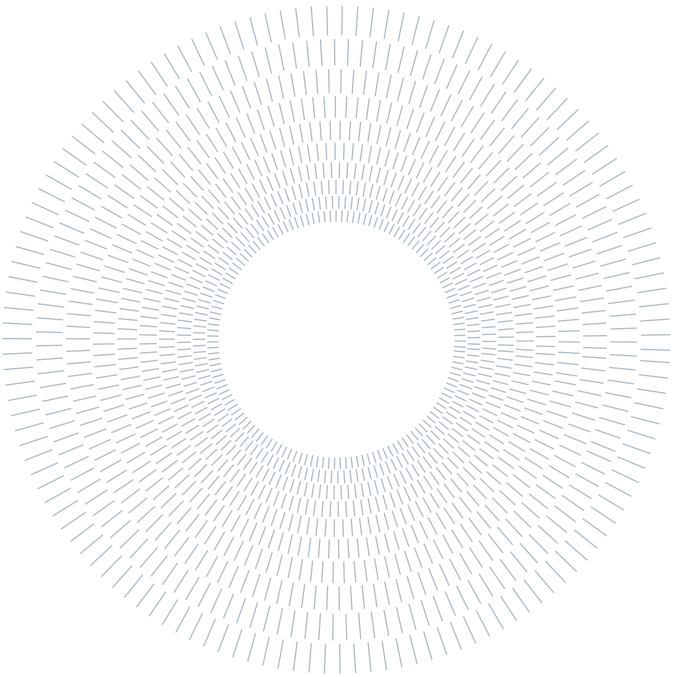


Figure 3.3: Login to BBP

UC3. Logout to BBP

Actor	abc
Entry conditions	abc
Event Flow	• abc
Exit condition	abc
Exceptions	• abc

Table 3.4: Logout to BBP

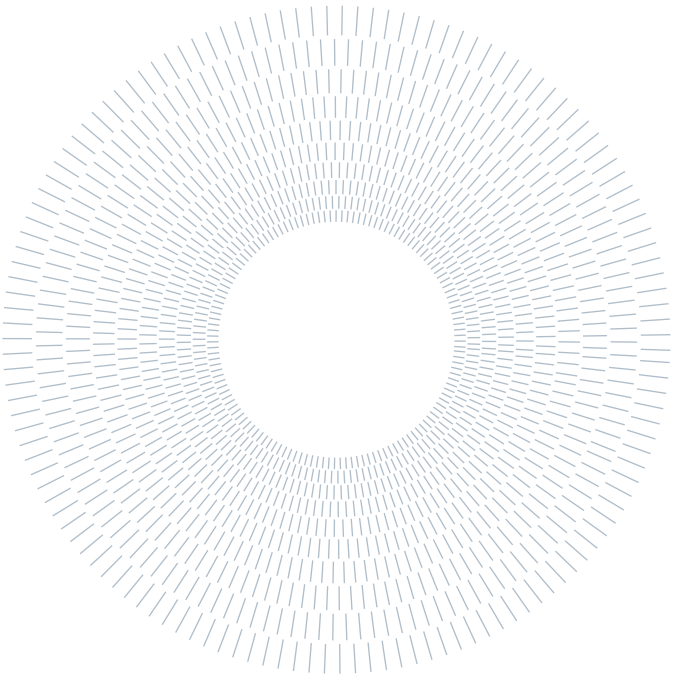


Figure 3.4: Logout to BBP

UC4. Recording with Manual Mode

Actor	abc
Entry conditions	abc
Event Flow	• abc
Exit condition	abc
Exceptions	• abc

Table 3.5: Recording with Manual Mode

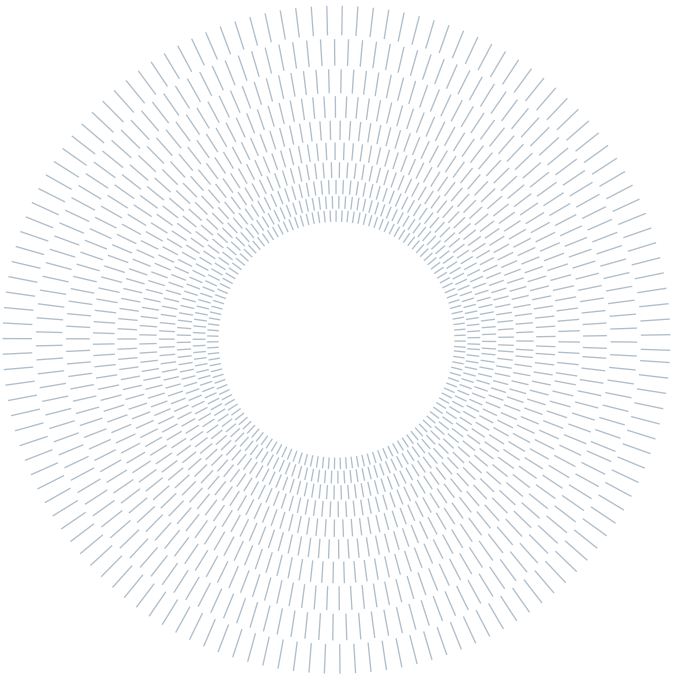


Figure 3.5: Recording with Manual Mode

UC5. Recording with Automated Mode

Actor	abc
Entry conditions	abc
Event Flow	• abc
Exit condition	abc
Exceptions	• abc

Table 3.6: Recording with Automated Mode

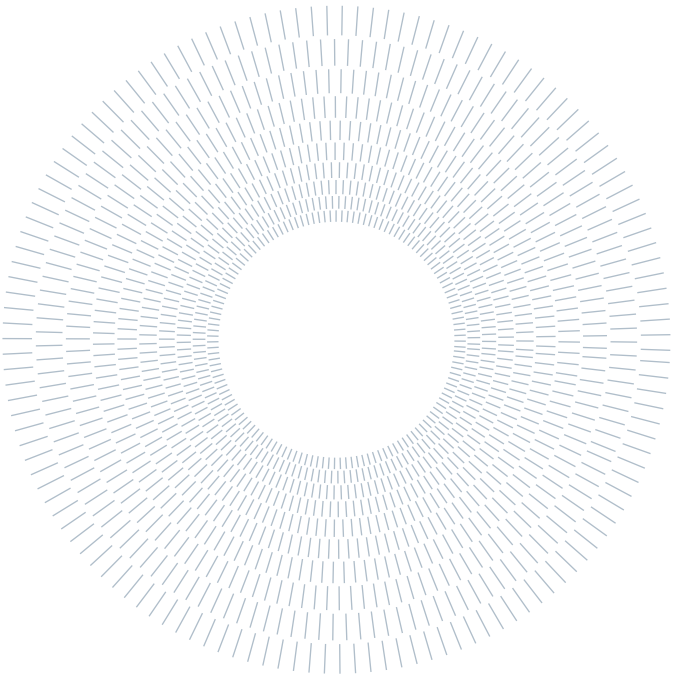


Figure 3.6: Recording with Automated Mode

UC6. Personal Trip Storage

Actor	abc
Entry conditions	abc
Event Flow	• abc
Exit condition	abc
Exceptions	• abc

Table 3.7: Personal Trip Storage

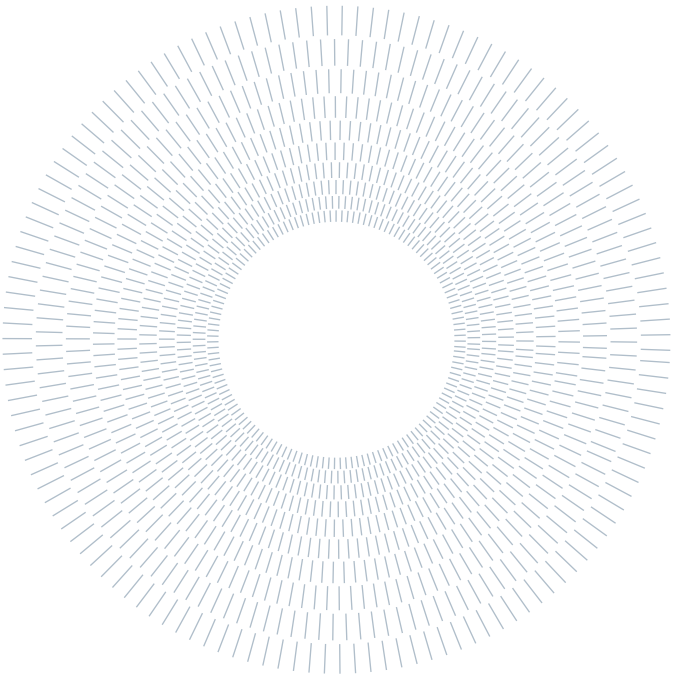


Figure 3.7: Personal Trip Storage

UC7. Search Path

Actor	abc
Entry conditions	abc
Event Flow	• abc
Exit condition	abc
Exceptions	• abc

Table 3.8: Search Path

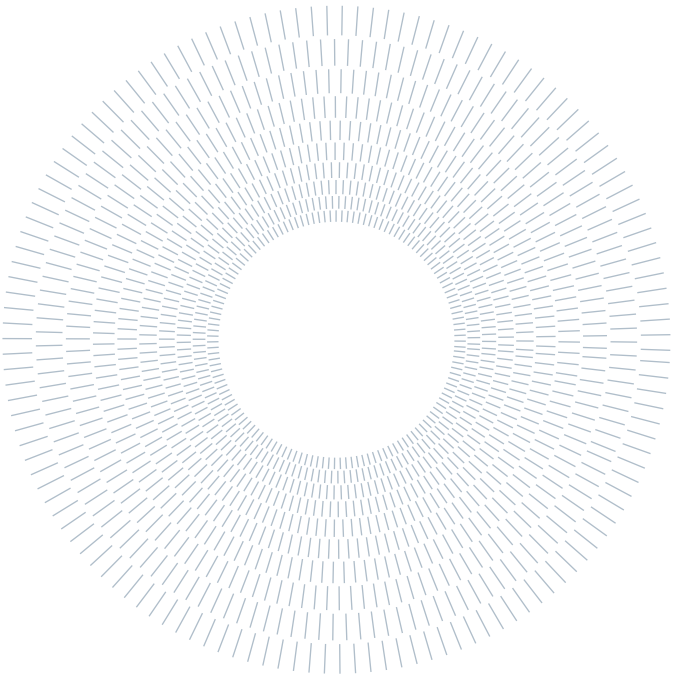


Figure 3.8: Search Path

UC8. Look Publish Path

Actor	abc
Entry conditions	abc
Event Flow	• abc
Exit condition	abc
Exceptions	• abc

Table 3.9: Look Publish Path

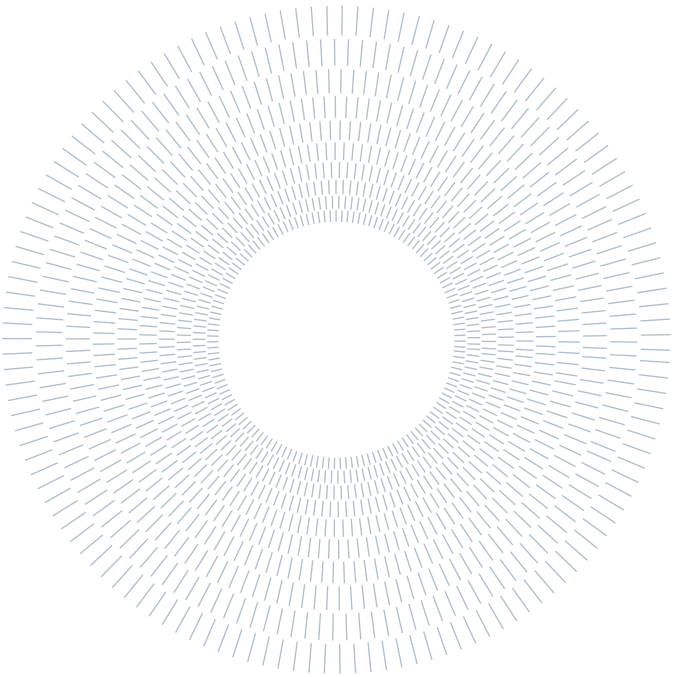


Figure 3.9: Look Publish Path

UC9. Account Settings

Actor	abc
Entry conditions	abc
Event Flow	• abc
Exit condition	abc
Exceptions	• abc

Table 3.10: Account Settings

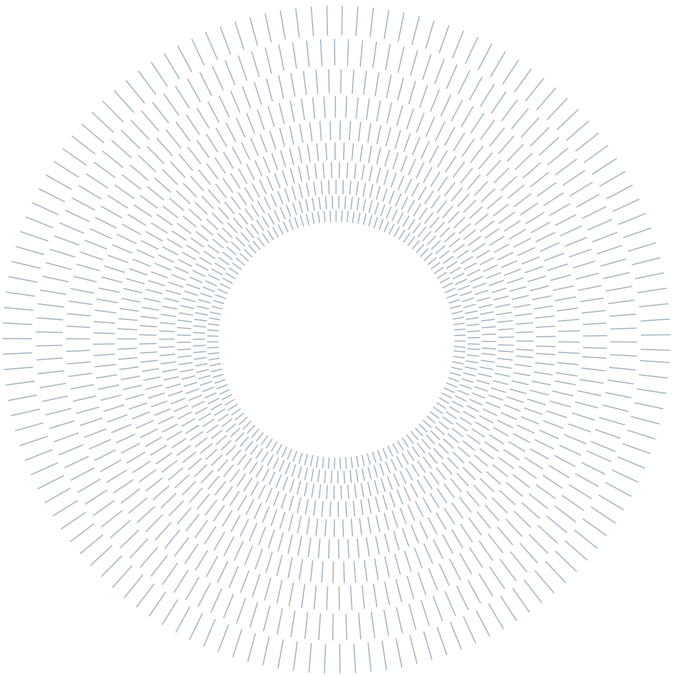


Figure 3.10: Account Settings

3.1.4. Mapping on Goals

G1 : Recognize suitable bike paths

- **R1:** abc
- **R2:** abc

G2 : Allow the user to browse different bike paths

- **R1:** abc
- **R2:** abc

G3 : Allow the user to see bike paths statistics

- **R1:** abc
- **R2:** abc

G4 : Allow the user to search for paths between two locations

- **R1:** abc
- **R2:** abc

G5 : Allow the user to see paths optimality based on other registered users' reviews

- **R1:** abc
- **R2:** abc

G6 : Allow user to see weather information when available

- **R1:** abc
- **R2:** abc

G7 : Allow users to register to the application using credentials

- **R1:** abc

- **R2:** abc

G8 : Allow registered user to review published bike paths

- **R1:** abc
- **R2:** abc

G9 : Allow registered user to confirm or reject false positives in track recording

- **R1:** abc
- **R2:** abc

G10 : Allow registered user to let the system automatically record suitable bike paths

- **R1:** abc
- **R2:** abc

G11 : Allow registered user to manually create suitable bike paths

- **R1:** abc
- **R2:** abc

G12 : Allow registered user to publish bike paths

- **R1:** abc
- **R2:** abc

G13 : Allow the store of bike paths data for registered users

- **R1:** abc
- **R2:** abc

G14 : Allow registered user to add publishable information about bike paths

- **R1:** abc
- **R2:** abc

Req.	G1	G2	G3	G4	G5	G6	G7	G8	G9	G10	G11	G12	G13	G14
R1	✓	✓												
R2	✓													
R3		✓												
R4	✓													
R5	✓													
R6			✓											
R7	✓													
R8	✓													
R9			✓											
R10	✓													
R11	✓													
R12			✓											
R13	✓													
R14	✓													
R15			✓											
R16	✓													
R17	✓													
R18			✓											
R19	✓													
R20	✓													
R21			✓											
R22	✓													
R23	✓													
R24			✓											
R25	✓													
R26	✓													
R27			✓											
R28	✓													
R29	✓													
R30			✓											

Req.	G1	G2	G3	G4	G5	G6	G7	G8	G9	G10	G11	G12	G13	G14
R31	✓													
R32	✓													
R33			✓											
R34	✓													
R35	✓													
R36			✓											
R37	✓													

Table 3.11: Mapping Requirements on Goals

3.2. Performance Requirements

The system relies heavily on real-time data acquisition, geospatial processing, and user interaction through mobile devices. Therefore, specific performance constraints must be met to ensure a proper user experience and the integrity of the collected data.

- **Response Time and Latency:** The system must ensure low latency during critical user interactions to maintain usability. Specifically, for general browsing and viewing of bike paths, the system should render map visualizations and path details within 3 seconds of a user's request, assuming a standard 4G/5G connection. For the route-finding feature, the algorithm computing the path score and optimal route must complete execution within 5 seconds. Regarding the "Automated mode," the system is required to process sensor inputs (accelerometer and gyroscope) and GPS data in real-time. The lag between the physical detection of an obstacle and its internal logging on the device should be very reduced to ensure accurate timestamping and location tagging. However, the complex merging of data from the same path can be performed asynchronously on the server side, provided that the updated status becomes available to the community within a reasonable timeframe like 5 minutes of data upload
- **Number of Concurrent Users:** As a community-driven platform managed by a biker's association, BBP must support a significant number of concurrent users, particularly during periods where the numbers of bikers could increase a lot, such as weekends and holidays. The system backend must be capable of handling at least 1,000 concurrent active sessions without performance degradation
- **Data Storage and Capacity:** The storage infrastructure must be designed to accommodate high-volume geospatial and time-series data. Since the application records continuous

GPS tracks, along with high-frequency accelerometer and gyroscope readings for obstacle detection, the database will grow rapidly. The system requires a scalable storage solution capable of archiving historical trip data while keeping active path inventories and recent statistics immediately accessible. Provisioning should be made for storing several Terabytes of data over the first few years of operation, with a strategy for data retention that prioritizes the freshness of path conditions

- **Network Bandwidth and Connectivity:** The system must be resilient to variable network conditions. The application should optimize data transmission to minimize bandwidth usage, which is crucial for users with limited data plans. In "Automated mode," the system must implement a "store-and-forward" mechanism: if the user enters an area with poor or no connectivity, the application must locally buffer all GPS and sensor data. Once connectivity is restored, the system should upload the accumulated data without freezing the user interface or interrupting the ongoing recording of the trip. The transmission of external meteorological data should be lightweight, fetching updates only at necessary intervals rather than continuously streaming, to preserve bandwidth
- **Energy Efficiency:** A critical performance requirement for the system's component is battery consumption. The simultaneous usage of the GPS module, accelerometer, gyroscope, and background processing algorithms constitutes a high energy drain. The application must be optimized to ensure that running the "Automated mode" doesn't reduce a standard smartphone battery excessively fast. The optimal target should be to allow a user to track a continuous ride of at least 5 to 8 hours on a fully charged device

3.3. Design Constraints

The design and development of the Best Bike Paths (BBP) system must adhere to specific boundaries imposed by the operational environment, regulatory requirements, and the technological limitations of the end-user devices. These constraints influence the architectural decisions and the selection of technologies to ensure the system is robust, legal, and usable.

3.3.1. Standard compliance

Given the nature of the application, which involves the collection, storage, and processing of user data and geolocation tracking, the system must strictly comply with relevant data protection regulations. Specifically, since the application tracks users' movements and associate trips with registered profiles, adherence to EU's GDPR (General Data Protection Regulation) is mandatory. Moreover, the system relies on external services. Therefore, it must comply with standard geospatial data formats to ensuring interoperability with standard mapping APIs and the navigation modules. The communication between the mobile application, the backend server, and the external meteorological service must adhere to modern security standards, necessitating the use of HTTPS and TLS protocols to encrypt data in transit. Finally, the integration of weather data implies compliance with the specific API standards provided by the chosen third-party meteorological service provider.

3.3.2. Hardware limitations

The "Automated mode" of the BBP application introduces significant hardware dependencies that act as constraints on the system's availability and performance. The primary limitation is the dependence on specific mobile sensors. For the application to function user's mobile device must be equipped with a high-precision GPS receiver, an accelerometer, and a gyroscope. Additionally, the continuous usage of these sensors imposes a constraint regarding battery consumption. The design must optimize sensor polling rates to prevent excessive battery drainage, which would negatively impact the user experience during long bike trips. Finally, the system is constrained by network connectivity hardware. While the application collects data locally, the features related to retrieving meteorological information, publishing path data, and merging crowd-sourced information require an active internet connection. The system design must, therefore, account for intermittent connectivity scenarios, typical of outdoor cycling routes, by implementing local caching mechanisms until a stable connection is re-established.

3.4. Software System Attributes

This section details the non-functional requirements that define the quality constraints of the Best Bike Paths system. These attributes are critical to ensuring the system is not only functional but also trustworthy, accessible, and sustainable over time.

3.4.1. Reliability

Given the nature of real-time app data acquisition and community-driven content, the reliability of the system is fundamental. The system must ensure high accuracy in its Automated Mode, specifically in the interpretation of sensor data to differentiate insignificant movements from truly obstacle. Furthermore, the data merging mechanism, which consolidates conflicting path statuses based on freshness, must be algorithmically robust. It must consistently produce a single, logical status for a path without data corruption, even when multiple users submit updates simultaneously on the same path. Finally, the system must handle the failure of the external API service gracefully, in general, the system must be fault tolerant to prevent the propagation of errors and to guarantee a continuous usability.

3.4.2. Availability

For the server-side components, the system generally targets a 99.9% uptime during daylight hours to support routing requests. Downtime for maintenance should be scheduled during hours of statistically low cycling activity to minimize disruption to the user base. A critical aspect of BBP is the tolerance regarding network connectivity, because if a biker is traveling a path with a poor or inexistent coverage the application must store the data until the connection is restored.

3.4.3. Security

Security is focused on two main aspects: Data Privacy and Information Integrity. The privacy for the system is important because the tracking of user movements, origin-destination points, and cycling habits, this geolocation data. All these data are highly sensitive. All personal data, particularly GPS logs, must be encrypted for safeness. Users' trip histories must be strictly accessible only to the account owner unless explicitly published. Instead, integrity must be ensured for data about path, score, obstacles and any other data useful for the usability of the system, they must be protected against malicious manipulation. The system should implement authentication protocols to ensure that only registered users can modify path statuses.

3.4.4. Maintainability

The architecture must be modular to facilitate future updates, error correction and scalability. Specifically, the component responsible for interfacing with mobile hardware (GPS, accelerometer, gyroscope) and the external meteorological service should be decoupled from the core business logic structure. This allows developers to update sensor calibration or adopt new hardware APIs without rewriting the trip management logic or change provider for obtaining weather information. The database schema should also be designed to be extensible, allowing for the potential addition of new metrics to the trip statistics in future versions.

3.4.5. Portability

Portability is a functional necessity for the client-side application. The BBP mobile app must be compatible with the major mobile operating systems like Android and iOS to reach a sufficiently large community of bikers. Possible future addition could be the creation of website for only help user to visualize / planning trips, in that case the system will be cross-browser compatible, functioning correctly on all modern web browsers.

4 | Formal Analysis Using Alloy

For readability and simplicity motivations we decided to insert the most important objects of the model simplifying the complex ones (for example, individual nodes of streets are not modeled) and we kept what we thought was essential to verify simple assertions.

4.1. Signatures

4.1.1. Status

```
abstract sig Status{}
one sig Clear extends Status{}
one sig RoadWork extends Status{}
one sig Pothole extends Status{}
```

Listing 4.1: Status signature

An object Status is used to be associated with a street segment, therefore allowing us to better describe streets and to introduce potholes. The actual signature is abstract since we wanted to make the object expandable with more variants (we added Clear, RoadWork and Pothole as examples).

4.1.2. Segment

```
sig Segment{
    nextSegment: lone Segment,
    status: some Status
}{
    // a segment cannot point to itself as next
    // (but cycles are allowed)
    nextSegment != this

    // if status is clear then its the only status possible
    Clear in status implies #status = 1
}
```

Listing 4.2: Segment signature

An object Segment is used to model street segments: every segment is connected to at most another segment (forming what actively is a chain of streets). To have consistent segments, they are to be considered as a part of a physical street and they also model direction of said street: therefore two different segments can be the same street segment but traversed in opposite direction, this implies that a necessary condition that we need to add is that a segment cannot connect to itself.

4.1.3. Path

```
sig Path{
    start: Segment,
    score: Int
}{
    // Paths are saved only if associated with
    // at least a Trip
    some t: Trip| t.path = this
}
```

Listing 4.3: Path signature

Following the definition of a Segment, a Path is simply an objects that keeps the first segment of an arbitrarily long chain of segments. Paths also have a score and, since paths are discovered and inserted by RegisteredUsers, it is necessary that a Trip (see Trip signature) has said Path associated to it.

4.1.4. Statistics

```
sig Statistics{}{
    // Statistics are saved only if associated
    // with at least a trip
    some t: Trip| t.averageSpeed = this
    or t.duration = this
    or t.externalStats = this
}
```

Listing 4.4: Statistics signature

4.1.5. Trip

```
sig Trip{
    path: one Path,
    averageSpeed: one Statistics,
    duration: one Statistics,
    externalStats: lone Statistics
}{
    // statistics are different
    averageSpeed != duration and duration != externalStats

    // Trips are saved only if associated with
    // at least a registered user
    one u: RegisteredUser| this in u.registeredTrips
}
```

Listing 4.5: Trip signature

A Trip is an object that has one single path, the statistics associated with the trip itself inserted by the user and the statistics retrieved from the external service. Because Trips are created by the user, it is necessary that every Trip has exactly one Path and that one RegisteredUser has registered said Trip.

4.1.6. RegisteredUser

```
sig RegisteredUser {
    registeredTrips: set Trip,
    publishedTrips: set Trip
}{
    // published trips are a subset of all registered trips
    publishedTrips in registeredTrips
}
```

Listing 4.6: RegisteredUser signature

A RegisteredUser models a user with ability to record trips. We decided not to model Guests in Alloy because the functionalities we wanted to test are used only by users with an account, so the users that we used have published trips and registered trips, where the published trips are naturally a subset of the registered ones.

4.2. Facts

Introduction (explain why it is needed that facts are to be inserted)

4.2.1. Segment facts

```
fact segmentsFacts{

    // Segments are saved only if associated with another
    // segment or a Path (and at least an object Path
    // is needed in the chain)
    all s: Segment| (
        some ss: Segment|
            ss.nextSegment = s  or
            some p: Path| p.start = s)

    all s: Segment|
        // if segment is last in chain or in a cycle
        (s.nextSegment = none or s in s.^nextSegment) implies
        // then from said segment i can walk backwards
        // until i find a path's start
        (some p: Path | p.start in s.^~nextSegment)
}
```

Listing 4.7: Segments fact

Explain why segments need more facts

4.2.2. Path facts

```
fact pathsFacts{
    // Paths are equal if the segments inside
    // are the same sequence of objects
    all p1,p2: Path|
        p1 != p2 implies (
            showPath[p1] != showPath[p2])
}
```

Listing 4.8: Paths fact

Explain why paths need more facts

5 Effort Spent

A table representing the total hours of work is shown here.

Work Areas	Alessandro Burato	Gabriele Crimella	Davide Faccini
Introduction	1.5h	1.5h	0h
Overall Description	0h	0h	0h
Specific Requirements	0h	0h	0h
Alloy	0h	0h	0h
Group Meetings and Reviews	8h	8h	8h

Table 5.1: Table of Effort

6 | References

Chapter 6

List of Figures

3.1 Use Cases 10
3.2 Sign in to BBP 11
3.3 Login to BBP 13
3.4 Logout to BBP 15
3.5 Recording with Manual Mode 17
3.6 Recording with Automated Mode 19
3.7 Personal Trip Storage 21
3.8 Search Path 23
3.9 Look Publish Path 25
3.10 Account Settings 27

List of Tables

1.1 List of Goals . . . 2
1.2 World Phenomena . . . 2
1.3 Machine Phenomena . . . 3
1.4 Shared Phenomena – World-Controlled . . . 3
1.5 Shared Phenomena – Machine-Controlled . . . 4

3.1 Requirements List . . . 9
3.2 Sign in to BBP . . . 9
3.3 Login to BBP . . . 12
3.4 Logout to BBP . . . 14
3.5 Recording with Manual Mode . . . 16
3.6 Recording with Automated Mode . . . 18
3.7 Personal Trip Storage . . . 20
3.8 Search Path . . . 22
3.9 Look Publish Path . . . 24
3.10 Account Settings . . . 26
3.11 Mapping Requirements on Goals . . . 32

5.1 Table of Effort . . . 41